Higher National Diploma in Information Technology

First Year, Second Semester Examination -2020

HNDIT1214- Statistics for IT

Question 01.

(i). Distinguish between descriptive statistics and inferential statistics. (4 marks)

(ii). What do you mean by sampling? Describe briefly main types of sampling methods. (6 marks)

(iii). The following are the number of replacement parts used in a factory in 50 consecutive weeks for a certain group of machines:

49	41	45	52	47	46	42	43	46	48
45	35	56	44	61	68	54	58	51	47
47	49	42	48	53	48	41	65	45	52
58	50	55	45	43	72	63	45	38	43
42	47	43	49	46	57	49	44	47	48

a) Construct a frequency distribution for this data by selecting class width as 5. (5 marks)

b) Represent the above data in a histogram and frequency polygon. (6 marks)

(iv). Write each of the following series in sigma notation.

a) $1 - \frac{1}{4} + \frac{1}{9} - \frac{1}{16} + \frac{1}{25} - \frac{1}{36}$ (2 marks)

b) $4^2 + 5^2 + 6^2 + 7^2 + \ldots + (n+2)^2$ (2 marks)

Question 02.

(i). What do you mean by "measure of central tendency"? (2 marks)

(ii). Following information represent the daily income of 150 families.

Income (Rs)	Number of families
more than 1000	150
more than 1500	140
more than 2000	115
more than 2500	95
more than 3000	65
more than 3500	53
more than 4000	32
more than 4500	15

a) Re arrange the above data as a frequency distribution (4 marks)

b) Calculate the following measurements for the above information

i) Mode (3 marks)

ii) Mean (3 marks)

iii) Variance (4 marks)

(iii). a) In a factory employing 3000 persons, 5% earn less than Rs.3 per hour, 580 earn from Rs. 3.01 to Rs. 4.50 per hour, 30% earn from Rs. 4.51 to Rs.6.00 per hour, 500 earn from Rs. 6.01 to Rs. 7.50 per hour, 20% earn Rs.7.51 to Rs.9.00 per hour, and the rest earn Rs. 9.01 or more per hour. What is the median wage? (4 marks)

b) The mean of five observations is 4.4 and the variance is 8.24. If three of the five observations are 1, 2 and 6, find the other two. (5 marks)

Question 03.

(1). If P = "Pr+1 and "C, = "Cr-1, then find the values of n and r. (4 marks)

(ii). Without repeating digits, from the digits 2, 3, 4, 5 and 6;

a) How many numbers greater than 4000 can be formed? (3 marks)

b) How many 4-digit numbers would be even? (3 marks)

(iii). A television director is scheduling a certain sponsor's commercials for an upcoming broadcast. There are six slots available for commercials. In how many ways may the director select the commercials;

a) If the sponsor has six different commercials, each to be shown once? (2 marks)

b) If the sponsor has three different commercials, each to be shown twice? (2 marks)

c) If the sponsor has three different commercials, the first of which is to be shown three times, the second two times, and the third once? (3 marks)

(iv). The word "ORDINATE" consists of eight letters, four vowels (O, I, A, E) and four consonants (R, D, N, T). Without repeating the letters form the words. Find;

a) The total number of words using all letters (2 marks)

b) The number of words starting and end with a vowel (3 marks)

c) The number of words in which all vowels are together. (3 marks)

Question 04.

(i). Define the term "probability". (2 marks)

(ii). In the history of play 12 games of chess, Carlsen wins 6. Levon wins 4 and two are drawn. They agree to play three games more. Find the probability that:

a) Carlsen wins all the three games (2 marks)

b) Two games end in a tie (3 marks)

c) Levon wins at least one game (3 marks)

(iii). A man wants to marry a girl having qualities: White complexion the probability of 1 getting such a girl is handsome dowry the probability of getting this is 20 westernized style the probability is 1 100 to such a girl, who has all the three qualities. 1 50 Find out the probability of his getting married to such a girl, who has all the three qualities. (3 marks)

(iv). A health club has three categories of membership: adult, social and junior. The number of members in each category, classified by gender, is shown in the table below.

	Adult	social	junior
Female	25	35	40
Male	95	25	80

One member is chosen, at random, to cut the ribbon at the opening of the new clubhouse.

a) Find the probability that:

I. a female member is chosen; (2 marks)
II. a junior member is chosen; (2 marks)
III. junior member is chosen, given that a female member is chosen. (3 marks)

b) V denotes the event that a female member is chosen W denotes the event that an adult member is chosen X denotes the event that a junior member is chosen For the events. V, W and X:

i. Write down two which are mutually exclusive (2 marks)

ii. Find two which are neither mutually exclusive nor independent. Justify your answer.

(3 marks)

Question 05.

(i). A random variable X has the following probability function:

X	-2	-1	0	1	2	3
P(X=x)	0.1	k	0.2	2k	0.1	3k

a) Find the following:

a. The value of k. (3 marks)

b. $P(x \geq 2)$ (2 marks)

c. $P(-2<x<2)$ (2 marks)

d. Expected value (3 marks)

(ii). The probability that a student must stop at any one traffic light coming to ATI is 0.2. There are 15 sets of traffic lights on the journey.

a) What is the probability that a student must stop at exactly 2 of the 15 sets of traffic lights? (3 marks)

b) What is the probability that a student will be stopped at one or more of the 15 sets of traffic lights?

(iii). A large group of students took a test in Statistics and the final marks have a mean of 70 and a standard deviation of 10. In that test, if a student obtains marks 60 or more he can pass the test otherwise fail. If we can approximate the distribution of these marks by a normal distribution, what percent of the students;

a) scored more than 80. (3 marks)

b) should pass the test. (3 marks)

c) should fail the test. (3 marks)